Chapter 4, Section 2

James Lee

August 11, 2026

1. Use (2.5.3) to show that $\mathbb{P}^n$ is simply connected.

Proof. Proceed by induction on n. The base case $n = 1$ is (2.5.3). Let $f : X \to \mathbb{P}^n$ be an étale covering. We may assume X is connected. Then X is smooth over k since f is étale (III, 10.1), and X is projective over k since f is finite, so X is a smooth irreducible projective variety of dimension n. It is enough to show the degree of f is 1. By (III, Ex. 11.3), $f^{-1}(H) = X \times_{\mathbb{P}^n} H$ is smooth and irreducible for almost every hyperplane $H \subset \mathbb{P}^n$. Also, the induced morphism $f' : f^{-1}(H) \to H$ is an étale covering of $H \cong \mathbb{P}^{n-1}$, since étale morphisms are stable under base change. By the inductive hypothesis, f' is an isomorphism. Hence, f is an isomorphism. □

2. *Classification of Curves of Genus 2.* Fix an algebraically closed field k of characteristic $\neq 2$.

(a) If X is a curve of genus 2 over k, the canonical linear system $|K|$ determines a finite morphism $f : X \to \mathbb{P}^1$ of degree 2 (Ex. 1.7). Show that it is ramified at exactly 6 points, with ramification index 2 at each one. Note that f is uniquely determined, up to an automorphism of $\mathbb{P}^1$, so X determines an (unordered) set of 6 points of $\mathbb{P}^1$, up to an automorphism of $\mathbb{P}^1$.

(b) Conversely, given six distinct elements $\alpha_1, \ldots, \alpha_6 \in k$, let K be the extension of $k(x)$ determined by the equation $z^2 = (x - \alpha_1)\cdots(x - \alpha_6)$. Let $f : X \to \mathbb{P}^1$ be the corresponding morphism of curves. Show that $g(X) = 2$, the map f is the same as the one determined by the canonical linear system, and f is ramified over the six points $x = \alpha_i$ of $\mathbb{P}^1$, and nowhere else.

(c) Using (I, Ex. 6.6), show that if P_1, P_2, P_3 are three distinct points of $\mathbb{P}^1$, then there exists a unique $\varphi \in \operatorname{Aut} \mathbb{P}^1$ such that $\varphi(P_1) = 0$, $\varphi(P_2) = 1$, $\varphi(P_3) = \infty$. Thus in (a), if we order the six points of $\mathbb{P}^1$, and then normalize by sending the first three to $0, 1, \infty$, respectively, we may assume that X is ramified over $0, 1, \infty, \beta_1, \beta_2, \beta_3$, where $\beta_1, \beta_2, \beta_3$ are three distinct elements of k, $\neq 0, 1$.

(d) Let Σ_6 be the symmetric group on 6 letters. Define an action of Σ_6 on sets of three distinct elements $\beta_1, \beta_2, \beta_3$ of k, $\neq 0, 1$, as follows: reorder the set $0, 1, \infty, \beta_1, \beta_2, \beta_3$ according to a given element $\sigma \in \Sigma_6$, then renormalize as in (c) so that the three become $0, 1, \infty$ again. Then the last three are the new $\beta_1', \beta_2', \beta_3'$.

(e) Summing up, conclude that there is a one-to-one correspondence between the set of isomorphism classes of curves of genus 2 over k, and triples of distinct elements $\beta_1, \beta_2, \beta_3$ of k, $\neq 0, 1$, modulo the action of Σ_6 described in (d). In particular, there are many non-isomorphic curves of genus 2. We say that curves of genus 2 depend on three parameters, since they correspond to the points of an open subset of $\mathbb{A}^3_k$ modulo a finite group.

Proof.

(a) By Hurtwitz's theorem, the degree of the ramification divisor is computed to be

$$\deg R = \deg K_X - 2 \deg K_{\mathbb{P}^1} = 6.$$

The ramification index at a point cannot exceed the degree of the morphism, so f must be ramified at exactly 6 distinct points each having ramification index 2.

(b) Let $B = k[x,z]/(z^2 - g(x))$ be an affine coordinate ring of X with $g(x) = (x-\alpha_1)\cdots(x-\alpha_6)$, and let f be the projection morphism onto the x-axis. By (II, 8.3A), we have

$$\Omega_{B/k} \cong (Bdx \oplus Bdz)/(g'(x)dx + 2zdz).$$

Let $P = (x', z')$ be a closed point in X. If $z' \neq 0$, then z is invertible in $\mathcal{O}_{P,X}$, so that

$$dz = -\frac{g'(x)}{2z}dx,$$

which means dx generates $(\Omega_{B/k})_P$. If $z' = 0$, then $x' = \alpha_i$ for some i, and $g'(x)$ is invertible in $\mathcal{O}_{P,X}$, so that

$$dx = -\frac{2z}{g'(x)}dz,$$

which means dz generates $(\Omega_{B/k})_P$. Since dx is a local parameter of $\mathbb{P}^1$, we see that P is ramified if and only if $z' = 0$ with ramification index $e_P = \nu_P(dx/dz) + 1 = 2$. There are six ramification points corresponding to the six roots of $g(x)$. Hence, $g(X) = 2$ by the Hurwitz formula.

$\square$

3. *Plane Curves.* Let X be a curve of degree d in $\mathbb{P}^2$. For each $P \in X$, let $T_P(X)$ be the tangent line to X at P (I, Ex.7.3). Considering $T_P(X)$ as a point of the dual projective plane $(\mathbb{P}^2)^*$, the map $P \to T_P(X)$ gives a morphism of X to its *dual curve* X^* in $(\mathbb{P}^2)^*$ (I, Ex. 7.3). Note that even though X is nonsingular, X^* in general will have singularities. We assume $\operatorname{char} k = 0$ below.

(a) Fix a line $L \subseteq \mathbb{P}^2$ which is not tangent to X. Define a morphism $\varphi : X \to L$ by $\varphi(P) = L \cap T_P(X)$, for each point $P \in X$. Show that φ is ramified at P if and only if either (1) $P \in L$, or (2) P is an *inflection point* of X, which means that the intersection multiplicity (I, Ex. 5.4) of $T_P(X)$ with X at P is $\geqslant 3$. Conclude that X has only finitely many inflection points.

(b) A line of $\mathbb{P}^2$ is a *multiple tangent* of X if it is tangent to X at more than one point. It is a *bitangent* if it is tangent to X at exactly two points. If L is a multiple tangent of X, tangent to X at the points $P_1, \ldots, P_r$, and if none of the P_i is an inflection point, show that the corresponding point of the dual curve X^* is an *ordinary r-fold point*, which means a point of multiplicity r with distinct tangent directions (I, Ex. 5.3). Conclude that X has only finitely many multiple tangents.

(c) Let $O \in \mathbb{P}^2$ be a point which is not on X, nor on any inflectional or multiple tangent of X. Let L be a line not containing O. Let $\psi : X \to L$ be the morphism defined by projection from O. Show that ψ is ramified at $P \in X$ if and only if the line OP is tangent to X at P, and that the ramification index is 2. Use Hurwitz's theorem and (I, Ex. 7.2) to conclude that there are exactly $d(d-1)$ tangents of X passing through O. Hence, the degree of the dual curve (sometimes called the *class* of X) is $d(d-1)$.

(d) Show that for all but a finite number of points of X, a point O of X lies on exactly $(d+1)(d-2)$ tangents of X, not counting the tangent at O.

(e) Show that the degree of the morphism φ of (a) is $d(d-1)$. Conclude that if $d \geqslant 2$, then X has $3d(d-2)$ inflection points, properly counted. (If $T_P(X)$ has intersection multiplicity r with X at P, then P should be counted $r-2$ times as an inflection point. If $r = 3$, we called it an *ordinary inflection point*.) Show that an ordinary inflection point of X corresponds to an ordinary cusp of the dual curve X^*.

(f) Now let X be a plane curve of degree $d \geqslant 2$, and assume that the dual curve X^* has only nodes and ordinary cusps as singularities (which should be true for sufficiently general X.) Then show that X has exactly $\frac{1}{2}d(d-2)(d-3)(d+3)$ bitangents.

(g) For example, a plane cubic curve has exactly 9 inflection points, all ordinary. The line joining any two of them intersects the curve in a third one.

(h) A plane quartic curve has exactly 28 bitangents. (This holds even if the curve has a tangent with four-fold contact, in which case the dual curve X^* has a tacnode.)

Proof.

(a) Find local affine coordinates x, y of $\mathbb{P}^2$ such that

$$X : y - f(x) = 0, \quad L : x = 0.$$

Setting x and y to be local parameters for X at P and L at $\varphi(P)$, respectively, we can write φ in the form

$$\varphi(x) = f(x) - xf'(x).$$

It follows that

$$\varphi^* dy = \frac{dy}{dx} dx, \quad \frac{dy}{dx} = -xf''.$$

By (2.2), φ is ramified at P if and only if $\nu_P(dt/du) > 0$, where we have

$$\nu_P\left(\frac{dy}{dx}\right) = \nu_P(x) + \nu_P(f'').$$

Notice that $\nu_P(x) > 0$ if and only if $P \in L$. Also, it is not hard to see

$$\mu_P(X, T_P(X)) = \nu_P(f'') + 2.$$

Thus, $\nu_P(dt/du) > 0$ if and only if $\nu_P(x) > 0$ or $\nu_P(f'') > 0$ if and only if (10)$P \in L$ or (2) P is an inflection point of X. Since $\varphi : X \to L$ ramifies at finitely many points, X has finitely many inflection points.

(b) Find local affine coordinates x, y of $\mathbb{P}^2$ such that

$$X : y - f(x) = 0, \quad L : y = 0.$$

This means x is a local parameter of X, so we can write $\gamma : X \ : X \to X^* \subset (\mathbb{P}^2)^*$ in the form

$$\gamma(x) = (-f'(x), xf'(x) - f(x)).$$

The tangent directions of $Q = \gamma(P_i)$ are precisely the images of the tangent vectors at P_i under the differential of γ. In particular, identifying the tangent vector of X at P_i to the row vector $(1, 0) \in T_{P_i}(X)$ and applying $d\gamma$ gives

$$v_i = (-f''(x_i), x_i f''(x_i)) \in T_Q(X^*).$$

We have shown in (a) that $f''(x_i) = 0$ if and only if P_i is an inflection point, which implies v_i and v_j define the same direction if and only if $P_i = P_j$. Hence, Q is an ordinary r-fold point with distinct tangent directions. Each multiple tangent of X corresponds to singular point of X^*, but a curve can have only finitely many singular points. Hence, X has finitely many multiple tangents.

(c) Find local affine coordinates x, y of $\mathbb{P}^2$ such that

$$X : y - f(x) = 0, \quad L : x = 0, \quad O = (1 : 0 : 0).$$

We can set $u = x$ and $t = y$ to be local parameters for X at P and L at $Q = \psi(P)$, respectively, and ψ is given by the projection morphism onto the x-axis. It follows that

$$\psi^* dy = \frac{dy}{dx} dx, \quad \frac{dy}{dx} = f'.$$

Hence, ψ is ramified at $P \in X$ if and only if $f'(P) = 0$ if and only if the tangent of X at P passes through O. Suppose ψ is ramified at P. Since O is not on any inflectional or multiple tangents of X, $\nu_P(f'') = 0$. Hence, $\nu_P(f') = \nu_P(f'') + 1 = 1$.

The genus of X is related to d by the degree genus formula. Since ψ has degree d as a morphism of curves, Hurwitz's theorem can be used to compute the degree of the ramification divisor

$$\deg R = \deg K_X - d \deg K_{\mathbb{P}^1} = d(d-1).$$

The ramification index of each ramification point is 2, so the degree of R equals to the number of tangents passing through O.

(d) This is another application of Hurwitz's theorem. The projection morphism $\psi : X - O \to L$ has degree $d-1$, so the degree of the ramification divisor of ψ is

$$\deg R = \deg K_X - (d-1)\deg K_{\mathbb{P}^1} = (d+1)(d-2).$$

A general point of X does not lie on any multiple tangents of X. Hence, the degree of R equals to the number of tangents passing through O.

(e) Define $\pi : (\mathbb{P}^2)^* - L \to L$ by $L' \mapsto L \cap L'$. Since φ factors into $\varphi = \pi \circ \gamma$, the degree of φ equals to the degree of the image of γ, which is $d(d-1)$. The degree of the ramification divisor of φ is

$$\deg R = \deg K_X - d(d-1)\deg K_{\mathbb{P}^1} = 3d^2 - 5d.$$

The degree of the ramification index also counts the number of points $X \cap L$. In particular, $X \cap L$ is generically d distinct points, and the ramification index of each $P \in X \cap L$ is 2, so we need to subtract d from $\deg R$ to get the total number of inflection points of X, which equals to $3d(d-2)$. If $r = 3$, an ordinary inflection point of X corresponds to a twisted cubic in the dual curve locally, which has an ordinary cusp.

(f) Let β_1 and β_2 be the number of bitangents and inflection points of X, respectively. We have two ways of computing the arithmetic genus of X^*: (1) in terms of $p_a(X)$, and (2) $X^* \subset (\mathbb{P}^2)^*$ as a plane curve of degree $d(d-1)$.

(1) The morphism $\gamma : X \to X^*$ explicitly defined in (b) is generically one-to-one by, so γ is a birational onto its image. The results of (Ex. 1.8) say

$$p_a(X^*) = p_a(X) + \delta, \quad \delta = \sum_{P \in X^*} \delta_P,$$

where $\delta_P = 1$ if $P \in X^*$ is a node or ordinary cusp, zero otherwise. In particular, $\delta = \beta_1 + \beta_2$ by (c) and (e).

(2) On the otherhand, X^* is a plane curve of degree $d(d-1)$ in $(\mathbb{P}^2)^*$. The degree-genus formula for plane curves (III, Ex. 4.7) gives

$$p_a(X^*) = \frac{1}{2}(d-2)(d+1)(d^2-d-1)$$

A smooth plane curve of degree d has $d(d-2)$ inflection points. Hence, the number of bitangets of X is

$$\beta_1 = p_a(X^*) - p_a(X) - \beta_2 = \frac{1}{2}d(d-2)(d-3)(d+3).$$

(g) Let $E \subset \mathbb{P}^2$ be a plane cubic curve, let $P, Q \in E$ be two inflection points, and let L the line through P and Q. If $P = Q$, then L is the tangent line through P. In this case, P is an inflection point, so it is counted with a multiplicity of 3, i.e., L meets no other point of X. Otherwise, L cannot be a tangent line for the same reason, and L meets X at a distinct point $R \neq P, Q$. The morphism $\varphi : E \to L$ defined in (a) is ramified at R. By (e), φ has degree $= 3(3-1) = 6$. Since P and Q are are ordinary inflection points, we have the ramification index of φ at P and Q is 2. Hence, R also has ramification index of 2, which is an inflection point.

□

5. *Automorphisms of a Curve of Genus* $\geqslant 2$. Prove the theorem of Hurtwitz [1] that a curve X of genus $g \geqslant 2$ over a field of characteristic 0 has at most $84(g-1)$ automorphisms. We will see later (Ex. 5.2) or (V, Ex. 1.11) that the group $G = \operatorname{Aut} X$ is finite. So let G have order n. Then G acts on the function field $K(X)$. Let L be the fixed field. Then the field extension $L \subseteq K(X)$ corresponds to a finite morphism of curves $f : X \to Y$ degree n.

(a) If $P \in X$ is a ramification point, and $e_p = r$, show that $f^{-1}f(P)$ consists of exactly n/r points, each having ramification index r. Let $P_1, \ldots, P_s$ be a maximal set of ramification points of X lying over distinct points of Y, and let $e_{P_i} = r_i$. Then show that Hurtwitz's theorem implies that

$$(2g-2)/n = 2g(Y) - 2 + \sum_{i=1}^{s}(1 - 1/r_i).$$

(b) Since $g \geqslant 2$, the left hand side of the equation is > 0. Show that if $g(Y) \geqslant 0$, $s \geqslant 0$, $r_i \geqslant 2$, $i = 1, \dots, s$ are integers such that

$$2g(Y) - 2 + \sum_{i=1}^{s}(1 - 1/r_i) > 0,$$

then the minimum value of this expression is $1/42$. Conclude that $n \leqslant 84(g-1)$.

Proof.

(a) Each $\sigma \in G$ induces an automorphism of X such that $f \circ \sigma = f$, so there is a group action of G on each fibre $f^{-1}f(P)$. The action is transitive (A.M., Ex. 5.13), and $e_P = e_{\sigma(P)}$ because σ induces an isomorphism of local rings $\mathcal{O}_{\sigma(P_i),X} \to \mathcal{O}_{P_i,X}$, hence sends uniformizer to uniformizer. In particular, if d_i is the size of the orbit of P_i, then $n = d_i r_i$. Hence, we have

$$\sum_{P \in X}(e_P - 1) = \sum_{i=1}^{s} \sum_{P \in f^{-1}f(P_i)} (e_P - 1) = \sum_{i=1}^{s} d_i(r_i - 1) = n\sum_{i=1}^{s}(1 - 1/r_i).$$

(b) The following give the minimum value

$$g(Y) = 0, \quad s = 3, \quad r_1 = \frac{1}{2}, \quad r_2 = \frac{1}{3}, \quad r_3 = \frac{1}{7}.$$

We conclude that

$$n \leqslant 42(2g-2) = 84(g-1).$$

$\square$

6. f_* *for Divisors.* Let $f : X \to Y$ be a finite morphism of curves of degree n. We define a homomorphism $f_* : \operatorname{Div} X \to \operatorname{Div} Y$ by $f_*(\sum n_i P_i) = \sum n_i f(P_i)$ for any divisor $D = \sum n_i P_i$ on X.

(a) For any locally free sheaf $\mathcal{E}$ on Y, of rank r, we define $\det \mathcal{E} = \bigwedge^r \mathcal{E} \in \operatorname{Pic} Y$ (II, Ex. 6.11). In particular, for any invertible sheaf $\mathcal{M}$ on X, $f_*\mathcal{M}$ is locally free of rank n on Y, so we can consider $\det f_*\mathcal{M} \in \operatorname{Pic} Y$. Show that for any divisor D on X,

$$\det(f_*\mathcal{O}_X(D)) \cong (\det f_*\mathcal{O}_X) \otimes \mathcal{O}_Y(f_*D).$$

Note in particular that $\det(f_*\mathcal{O}_X(D)) \neq \mathcal{O}_Y(f_*D)$ in general!.

(b) Conclude that f_*D depends only on the linear equivalence class of D, so there is an induced homomorphism $f_* : \operatorname{Pic} X \to \operatorname{Pic} Y$. Show that $f_* \circ f^* : \operatorname{Pic} Y \to \operatorname{Pic} Y$ is just multiplication by n.

(c) Use duality for a finite flat morphisms (III, Ex. 6.10) and (III, Ex. 7.2) to show that

$$\det f_*\Omega_X \cong (\det f_*\mathcal{O}_X)^{-1} \otimes \Omega_Y^{\otimes n}.$$

(d) Now assume that f is separable, so we have the ramification divisor R. We define the *branch divisor* B to be the divisor f_*R on Y. Show that

$$(\det f_*\mathcal{O}_X)^2 \cong \mathcal{O}_Y(-B).$$

Proof.

(a) Let P be any point in X, and let $f_*P = f(P)$ be the image of P. The skyskraper sheaf $k(f_*P)$ has a natural free resolution

$$0 \longrightarrow \mathcal{O}_Y(-f_*P) \longrightarrow \mathcal{O}_Y \longrightarrow k(f_*P) \longrightarrow 0.$$

In particular, $\det k(f_*P) \cong \mathcal{O}_Y(f_*P)$, and any finite free resolution of $k(f_*P)$ induces isomorphic determinant line bundles (II, Ex. 6.11). Also, the sheaves $k(f_*P) \to f_*k(P)$ are isomorphic, so any free resolution of $f_*k(P)$ is also a free resolution of $k(f_*P)$.

Now, let D be any non-zero divisor, and let P be any point. We will show that the formula is true for D if and only if it is true for $D+P$ (1.3). Consider the exact sequence

$$0 \longrightarrow \mathcal{O}_X(-P) \longrightarrow \mathcal{O}_X \longrightarrow k(P) \longrightarrow 0.$$

Tensoring with $\mathcal{O}_X(D+P)$ and applying f_* gives

$$0 \longrightarrow f_*\mathcal{O}_X(D) \longrightarrow f_*\mathcal{O}_X(D+P) \longrightarrow f_*k(P) \longrightarrow 0.$$

By the remarks above, we have an isomorphism

$$\mathcal{O}_Y(f_*P) \cong \det(f_*\mathcal{O}_X(D+P)) \otimes (\det(f_*\mathcal{O}_X(D)))^{-1},$$

which is what we wanted to show.

(b) By (a), since $f_*\mathcal{O}_X(D)$ depends only on the linear equivalence class of D, so does $\mathcal{O}_Y(f_*D)$. For the second part, it suffices to show the statement holds for any point $Q \in Y$. Indeed, it follows by definition $f_* \circ f^*$ is multiplication by some integer. On the other hand, f^* multiplies the degree by n (II, 6.9), and f_* preserves degrees. Hence, $f_* \circ f^*$ must be multiplication by n.

(c) Let $\mathcal{F}$ be a coherent sheaf on Y. By definition, we have

$$f_*f^!\mathcal{F} \cong f_*\mathcal{O}_X^{-1} \otimes \mathcal{F}, \quad f_*\mathcal{O}_X^{-1} = \mathcal{H}om_Y(f_*\mathcal{O}_X, \mathcal{O}_Y).$$

By (III, Ex. 7.3), $f^!\Omega_Y \cong \Omega_X$ is a dualizing sheaf on X. Applying above to $\mathcal{F} = \Omega_Y$ gives

$$\begin{aligned} \det f_*\Omega_X &\cong \det f_*f^!\Omega_Y \\ &\cong \det\big(f_*\mathcal{O}_X^{-1} \otimes \Omega_Y\big) \\ &\cong (\det f_*\mathcal{O}_X)^{-1} \otimes \Omega_Y^{\otimes n}. \end{aligned}$$

(d) Let K_X and K_Y be canonical divisors of X and Y, respectively. They satisfy the following relation (2.3)

$$f^*K_Y - K_X \sim -R.$$

Applying f_* gives

$$nK_Y - f_*K_X \sim -B.$$

On the otherhand, (a) applied to $D = K_X$ gives

$$\mathcal{O}_Y(-f_*K_X) \cong (\det f_*\mathcal{O}_X) \otimes (\det f_*\Omega_X)^{-1}.$$

and by (c), we have the isomorphism

$$\mathcal{O}_Y(nK_Y) \cong (\det f_*\mathcal{O}_X) \otimes (\det f_*\Omega_X),$$

Combining these two isomorphisms gives

$$(\det f_*\mathcal{O}_X)^2 \cong \mathcal{O}_Y(nK_Y - f_*K_X) \cong \mathcal{O}_Y(-B).$$

□

7. *Étale Covers of Degree 2.* Let Y be a curve over a field k of characteristic $\neq 2$. We show there is a one-to-one correspondence between finite étale morphisms $f: X \to Y$ of degree 2, and 2-torsion elements of $\operatorname{Pic} Y$, i.e., invertible sheaves $\mathcal{L}$ on Y with $\mathcal{L}^2 \cong \mathcal{O}_Y$.

(a) Given an étale morphism $f: X \to Y$ of degree 2, there is a natural map $\mathcal{O}_Y \to f_*\mathcal{O}_X$. Let $\mathcal{L}$ be the cokernel. Then $\mathcal{L}$ is an invertible sheaf on Y, $\mathcal{L} \cong \det f_*\mathcal{O}_X$, and so $\mathcal{L}^2 \cong \mathcal{O}_X$ by (Ex. 2.6). Thus an étale cover of degree 2 determines a 2-torsion element in $\operatorname{Pic} Y$.

(b) Conversely, given a 2-torsion element $\mathcal{L}$ in $\operatorname{Pic} Y$, define an $\mathcal{O}_Y$-algebra structure on $\mathcal{O}_Y \oplus \mathcal{L}$ by $\langle a, b\rangle \cdot \langle a', b'\rangle = \langle aa' + \varphi(b \otimes b'), ab' + a'b\rangle$, where φ is an isomorphism of $\mathcal{L} \otimes \mathcal{L} \to \mathcal{O}_Y$. Then take $X = \mathcal{S}pec(\mathcal{O}_Y \oplus \mathcal{L})$. Show that X is an étale cover of Y.

(c) Show that these two processes are inverse to each other.

Proof.

(a) We check that $\mathcal{L}$ is an invertible sheaf on Y. At the level of stalks, the natural map $\mathcal{O}_{Q,Y} \to (f_*\mathcal{O}_X)_Q \cong \mathcal{O}_{Q,Y}^{\oplus 2}$ is the diagonal map, so the cokernel is free of rank one. Hence, $\mathcal{L}$ is locally free of rank one.

(b) By construction, $f : X \to Y$ is a finite morphism, and $f_*\mathcal{O}_X \cong \mathcal{O}_Y \oplus \mathcal{L}$ is locally free of rank 2, so f is a flat morphism. Also, φ is locally given by multiplication, i.e., $\varphi(b \otimes b') = bb'$. If P is a point in Y, then there is an isomorphism

$$(f_*\mathcal{O}_X)_P \cong \mathcal{O}_{P,X}[\varepsilon]/(\varepsilon^2 - 1) \cong \mathcal{O}_{P,X} \times \mathcal{O}_{P,X},$$

so f is unramified at every $P \in Y$. Hence, $f : X \to Y$ is a degree 2 étale cover of Y.

(c) One direction is obvious. Consider the pullback diagram

$$\begin{tikzcd} X \times_Y X \arrow[r] \arrow[d] & X \arrow[d] \\ X \arrow[r, "f"] & Y \end{tikzcd}.$$

The projection morphisms $p_1, p_2 : X \times_Y X \to X$ are étale covers of degree 2. Also, the diagonal $\Delta \subset X \times_Y X$ is a closed subscheme of $X \times_Y X$ is isomorphic to X. Thus, the restriction $p_1, p_2 : X \times_Y X - \Delta \to X$ is an étale cover of X of degree 1, which is an isomorphism. Hence, $\tau = p_2 \circ p_1^{-1} : X \to X$ is an automorphism of X over Y which tranposes the fibres of f.

Now, let $f : X \to Y$ be a étale cove of degree 2, and let $X' = \mathcal{S}pec(\mathcal{O}_Y \oplus \mathcal{L})$, where $\mathcal{L}$ is the cokernel of the natural morphism $\mathcal{O}_Y \to f_*\mathcal{O}_X$. We want to show X and X' are isomorphic. It suffices to show the following short exact sequence of $\mathcal{O}_Y$-modules splits

$$0 \longrightarrow \mathcal{O}_Y \longrightarrow f_*\mathcal{O}_X \longrightarrow \mathcal{L} \longrightarrow 0.$$

The automorphism $\tau : X \to X$ of X over Y, which tranposes the fibres of f, induces an $\mathcal{O}_Y$-module automorphism $\tau_* : f_*\mathcal{O}_X \to f_*\mathcal{O}_X$. The kernel of the trace morphism $t = 1 - \tau : f_*\mathcal{O}_X \to f_*\mathcal{O}_X$ is exactly $\mathcal{O}_Y$. Hence, t induces a section $\mathcal{L} \to f_*\mathcal{O}_X$, so the short exact sequence splits. □